MIL-T-52340A(ME)
8 January 1974
SUPERSEDING
MIL-T-52340(MO)
20 December 1963

MILITARY SPECIFICATION

TOOL, VERTICAL CONNECTING: ERECTION EQUIPMENT FOR BRIDGE, FLOATING, PNEUMATIC FLOAT, CLASS 60

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a vertical connecting tool to be used as an erection aid in the construction of the class 60 floating bridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-S-200	- Set Screws: Hexagon Socket and Spline Socket, Headless.
RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-T-60	- Tape: Packaging, Waterproof.

FSC 5420

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Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Testing.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

C7940-1	- Tool, Vertical Connecting: Erection Equipment for Bridge, Floating, Pneumatic Float, Class 60.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

A36 - Structural Steel.

A441 - High-Strength, Low-Alloy Structural Manganese Vanadium Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.

A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.

B3.0 - Standard Qualification Procedure.

D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street N.W., Washington, DC 20036.)

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3. REQUIREMENTS

3.1 Description. The vertical connecting tool shall be as shown on Drawing C7940-1 and as specified herein.

3.1.1 Drawings. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a vertical connecting tool for inspection within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods will produce vertical connecting tools that comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the supplier and shall be subject to all provision of this specification.

3.3.1 Structural steel. Structural steel shall conform to ASTM A36.

3.3.2 Low-alloy structural steel. Low-alloy structural steel shall conform to ASTM A441.

3.4 Hydraulic ram. The hydraulic ram shall have a capacity of 7 tons with an internal pressure between 5900 and 6700 psi. The materials used in the hydraulic ram shall be compatible with fluid conforming to MIL-H-6083 or MIL-H-5606. When shipped, the hydraulic ram shall not be filled with hydraulic fluid except as specified in 5.1.1 and 5.1.2.

3.5 Hydraulic couplings. A set of hydraulic couplings shall consist of a female section for attachment to the ram: male section for attachment to a hydraulic hose; swivel of female section; and dust caps attached to the male and female sections with steel chain conforming to RR-C-271, type II, class 3, weldless, sash chain, zinc coated. The

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coupling components shall be capable of withstanding a minimum of 10,000 psi internal working pressure without any evidence of leakage or failure. All coupling parts shall be of corrosion-resistant material or shall be rendered corrosion resistant by chromium or cadmium plating. The coupling shall be of such design that breaking the connection at the swivel will cause spring-loaded devices to seal the fluid passage in each half coupling. The spring-loaded devices shall open to permit the fluid flow when a connection is made. The exterior of the coupling shall be shaped to provide a wrench grip. The swivel shall turn freely by hand.

3.6 Set screw. Set screws shall conform to FF-S-200, type I, style 1.

3.7 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required.

3.8 Identification marking. The vertical connecting tool shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.9 Treatment and painting. The vertical connecting tool, except for the couplings, screws, and threads, shall be treated painted in accordance with MIL-T-704, type B except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Both the inside and the outside surfaces of the vertical connecting tool weldment shall be painted. Screws and threads shall be coated with preservative conforming to MIL-C-16173, grade 1.

3.10 Workmanship.

3.10.1 Steel fabrication. Steel used in the fabrication of the vertical connecting tool shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

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3.10.2 Welders and welding.

3.10.2.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).

Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.10.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1 - Structural Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction vertical connecting tool shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The preproduction vertical connecting tool shall be tested as specified in 4.5.2. Failure to pass the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The vertical connecting tools shall be examined as specified herein, for the following defects:

Major

101. Nonconformance to the dimensions shown on the drawing.
102. Components missing or not as specified.
103. Material not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

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Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.

4.5.2 Performance test. Place the vertical connecting tool in a test frame as shown in figure 1. Connect a 10,000-psi-minimum pump to the vertical connecting tool incorporating the coupling set. Install a 10,000-psi-minimum-capacity gage in the hydraulic line between the pump and tool. Pump hydraulic fluid into the tool until the gage in the line indicates the pressure required to produce 7 tons. Maintain this pressure on the gage for 5 minutes. Then release the pressure and disconnect the couplings at the swivel. Determine whether there is leakage through or around the seal. Then connect the couplings again. Evidence of any of the following shall constitute failure of this test:

(a) Failure of or damage to the vertical connecting tool.
(b) Any leakage from the couplings.
(c) Leakage through or around the seal when the couplings are disconnected at the swivel.
(d) Any obstruction of the hydraulic fluid flow when the couplings are reconnected.
(e) Any leakage of hydraulic fluid or evidence of damage to the hydraulic ram.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

106. Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.

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107. Interior surfaces of the hydraulic ram not preserved as specified for level A or B.
108. Openings not sealed as specified for level A or B.
109. Contents not immobolized as specified for level A or B.
110. Box not waterproof sealed as specified for level A.
111. Quantities packed together exceed the weight or size limitations for level A or B.
112. Strapping not zinc coated for level A.
113. Marking missing, illegible, incorrect, or incomplete for level A, B or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Interior surfaces of the hydraulic ram shall be coated with the hydraulic fluid recommended for operation and the excess fluid drained. Openings into the ram shall be sealed with tape conforming to PPP-T-60, type IV. The tool with couplings shall be packaged in a close fitting box conforming to PPP-B-636, W5c. The contents shall be immobilized within the box in a manner to prevent free movement or damage by the use of fiberboard inserts, pads, or cushions, fabricated from the same material as the box. The box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

5.1.2 Level B. Each tool with couplings shall be preserved and packaged as specified in 5.1.1 for level A except the box and the inserts, pads, or cushions used shall conform to PPP-B-636, type CF, class domestic, variety SW, grade 175, style optional. Box closure shall be in accordance with the appendix to the box specification.

5.1.3 Level C. Each tool with couplings shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Tools preserved and packaged as specified in 5.1, shall be packed together in a close fitting box conforming to PPP-B-601,

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overseas type, style optional in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. Tools preserved and packaged as specified in 5.1, shall be packed together as specified in 5.2.1 except boxes shall be domestic type and strapping need not be zinc coated. As an alternate the tools may be packed in a close fitting box conforming to PPP-B-640, class 2, style optional or PPP-B-636, V3c, V11c, V13c, or V15c, style optional. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Tools preserved and packaged as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in a container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vertical connecting tool is to be used to vertically aline the deck sections of the class 60 floating bridge.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it has been determined that the tools will be held in covered storage.

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6.3 Preproduction model. Any changes or deviations of production vertical connecting tools from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish vertical connecting tools conforming to this specification.

Custodian:

Army - ME

Preparing activity:

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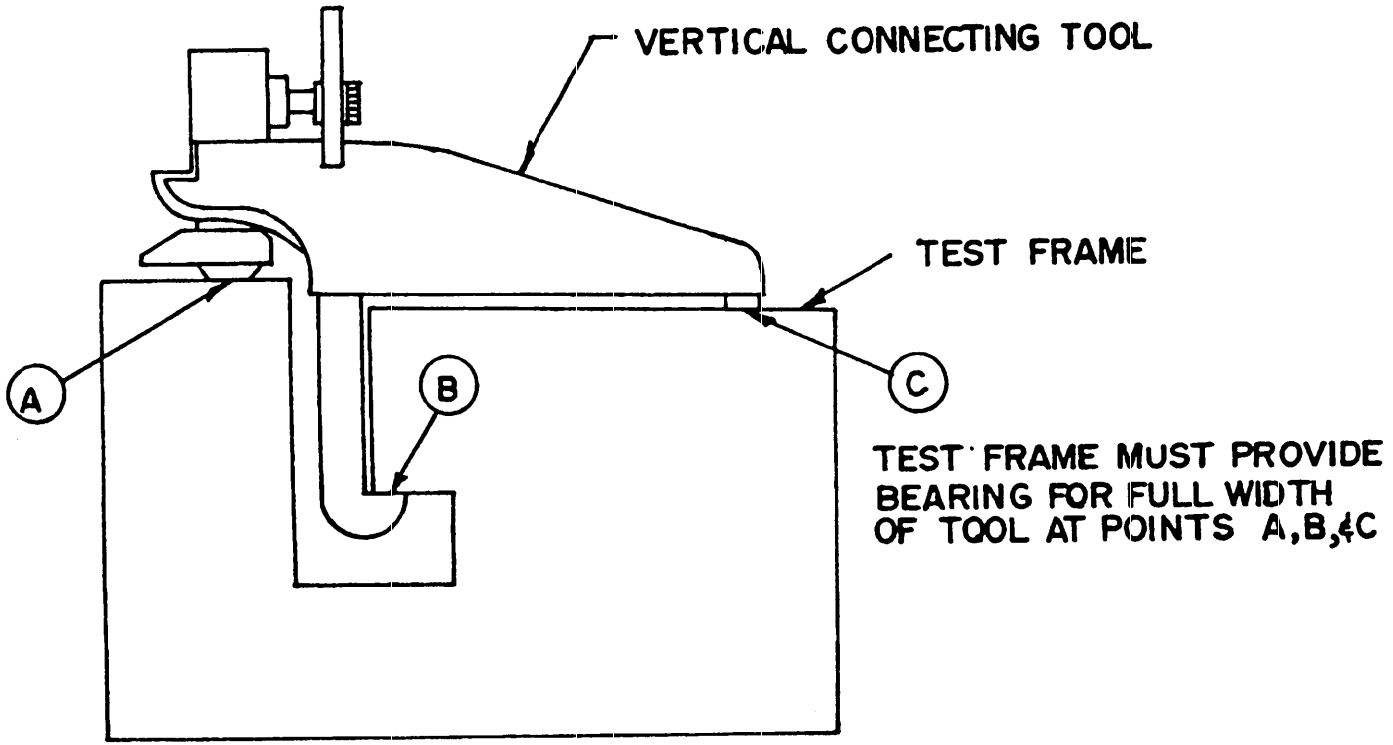


FIGURE 1. VERTICAL CONNECTING TOOL TEST FRAME.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-52340A(ME) Tool, Vertical Connecting: Erection Equipment for Bridge, Floating, Pneumatic Float, Class 60

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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